Journal Pre-proofs

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PII: S0960-8524(20)31435-8
DOI: https://doi.org/10.1016/j.biortech.2020.124161
Reference: BITE 124161

To appear in: *Bioresource Technology*

Received Date: 15 August 2020
Revised Date: 17 September 2020
Accepted Date: 18 September 2020

Recent advancement in remediation of synthetic organic antibiotics from environmental matrices: Challenges and Perspective

Preeti Chaturvedi[1, 2, #,*], Balendu Shekher Giri[1, #], Parul Shukla[#, 1], Pratima Gupta[2]

[1]Aquatic Toxicology Laboratory, Environmental Toxicology Group, Council of Scientific and Industrial Research-Indian Institute of Toxicology Research (CSIR-IITR), Vishvigyan Bhawan, 31, M.G. Marg, Lucknow 226001, Uttar Pradesh, India

[2]Department of Biotechnology, National Institute of Technology-Raipur, G.E. Road, Raipur, 492010, Chhattisgarh, India

Abstract

Continuous discharge and persistence of antibiotics in aquatic ecosystem is identified as emerging environment health hazard. Partial degradation and inappropriate disposal induce appearance of diverse antibiotic resistant genes (ARGs) and bacteria, hence their execution is imperative. Conventional methods including waste water treatment plants (WWTPs) are found ineffective for the removal of recalcitrant antibiotics. Therefore, constructive removal of antibiotics from environmental matrices and other alternatives have been discussed. This review summarizes present scenario and removal of micro-pollutants, antibiotics from environment. Various strategies including physicochemical, bioremediation, use of bioreactor, and biocatalysts are recognized as potent antibiotic removal strategies. Microbial Fuel Cells (MFCs) and biochar have emerged as promising biodegradation processes due to low cost, energy efficient and environmental benignity. With higher removal rate (20-50%) combined/ hybrid processes seems to be more efficient for permanent and sustainable elimination of reluctant antibiotics.

Keywords: Antibiotics; Bioreactors; Waste-water, Synthetic organic compounds

*** Corresponding author,** [#] contributed equally

1. Introduction

Occurrence and environmental fate of antibiotics has rapidly studied due to several virulent and hazardous effects associated with it. Although the antibiotic concentration in environment is found to be low, in ground water and wastewater at ng/L to μg/L, in soil and sludge at μg/kg to mg/kg, but their prolonged presence in environment can cause the emergence of highly resistant and transformed antibiotic resistant genes and bacterial strains (Li and Zhang, 2010). These antibiotics are capable of exerting pressure in transforming resistant bacteria. Also increased and exponential occurrence of clinical infections has been identified as major health issue and thus its remediation has been thought as pivotal task by many environmental researchers. World Health Organization (WHO), World Health Assembly, and European Centre for Disease Control have highlighted the AMR as a critical public health threat. World Health Organization (WHO) has proposed certain regional strategies to reduce the mortality and morbidity rate cause by resistant infections to preserve the beneficial effects of antimicrobial drugs. Therefore, there is need to investigate the long and short-term effect of antibiotics on human health and surroundings.

Global use of antibiotics has exceedingly enhanced in recent years, antibiotics of pharmaceutical origin can be easily found in surroundings. Consumption of antibiotics in livestock has been estimated at 63,151 tons in 2010, which is predicted to increase by 67% till 2030 (Kraemer et al., 2019). Sources like industries, effluent treatment plant, augmentation of antibiotics in animals and human medicine, other commercial uses and thus presence in sewage, has led all-time high level of antibiotics in aquatic environment (Álvarez et al., 2010). Many antibiotics that are hydrophilic in nature can be transported with running water and can spread to various aquatic bodies. Moreover, consumed antibiotics are not completely metabolized and excreted from the body in more or less same chemical form. It is approximated that the typical WWTPs are also unable to eliminate the antibiotics present in sewage and waste water.

Antibiotics are the chemotherapeutic agents, isolated from natural sources or synthetically synthesized and are used for the purpose of restraining microbial growth. These chemical substances are used to eradicate the bacterial infections via bactericidal or bacteriostatic mechanism. Antimicrobial agents affect viability of microbes by targeting microbial nucleic acid and protein synthesis, cell wall, cell membrane, and biological metabolic compound synthesis.

However, over usage and unprocessed disposal of antibiotics in today's world has given birth to new health related emergencies not just in developing countries but throughout the world. Apart from other chemical pollution, untreated antibiotics in water bodies can accelerate the generation of Antimicrobial Resistance Genes (ARGs). These ARGs promotes the emergence of Multi Drug Resistant (MDR) bacteria, which shades health risk to animals and humans. Microbes tend to aggregate and form biofilms, which prevents the penetration of antibiotics with the presence of matrix, acting as barrier for entry of antibiotics (M. Kumar et al., 2019). Negatively charged biofilm has investigated that was preventing the penetration of positively charged aminoglycosides antibiotics (streptomycin, gentamycin) (Radlinski and Conlon, 2018). In addition, mutations in genes, carrying resistant genes and plasmids, development of efflux pump mechanism, transfer of resistant genes carried by insertion sequences and transposons, and conjugation among species has altered the antimicrobial effect on pathogenic bacteria. Due to these exercises even the fourth-generation antibiotics has found to be resistant.

Risks of toxicity due to long-term exposure of antibiotics are still under investigation and may be associated with deep influence on terrestrial and aquatic ecosystem and their flora and fauna. Studies performed on effect of antibiotics have demonstrated that such organic substances even at low concentration (ngL^{-1}) can affect the reproduction and growth of birds, fishes, invertebrates, plants and bacteria (de Cazes et al., 2014). In addition, trace elements present in soil and water also affects human health which is directly related to food and agriculture.

Presence of prescribed and non-prescribed drugs such as beta-lactams, amoxicillin, ciprofloxacin, sulfamethoxazole, cetirizine, terbinafine, norfloxacin, citalopram, ibuprofen, carbamazepine, enoxacin, and many more have been detected at higher concentration in Indian aquatic system (Balakrishna et al., 2017).Various conventional techniques employed are found insignificant for the complete removal of antibiotics and their metabolites. Thus, implementation of other alternatives was considered. For instance, enzyme catalysis via Glutathion S-Transferases (GSTs) have been demonstrated on degradation of hydrophobic (having electrophilic centers) antibiotics including ampicillin, sulfathiazole, and tetracycline (Park, 2012).

This review deals with the various problems and factors occurred during utilization and application of numerous biological and non-biological processes for antibiotic treatment. For this purpose, many biodegradation treatments (aerobic and anaerobic) have been reviewed to study the interaction between antibiotics and biological methods. Degradation strategies such as Oxidative Degradation, adsorption, hydrolysis, bio-electrochemical methods, metal-assisted photolysis, phytolysis, membrane separation, coagulation and certain biodegradation processes i.e. use of bacterial, fungal, algae strains, and biocatalysts are investigated and have shown the possibility of antibiotic removal to a more nontoxic extent. Benefits associated with biochar and MFCs to degrade antibiotics have been investigated. Hence, this review presents the novelty in terms of detailed information provided about the various remediation processes along with their pros and cons. Moreover, future scope and potential of biodegradation over physicochemical methods and application of hybrid processes have been discussed.

1.2 Effect of antibiotics and present scenario

Increased quantity of antibiotics in aquatic environment may lead to many adverse effects on ecosystem and putrefaction of aquatic resources, thus developing constructive strategies for their degradation is imperative. Exposure of such chemical compounds (antibiotics) and their products, contribute to multidrug resistant (MDR) bacteria, ultimately diminishing the effectiveness of antibiotics. Resistance emergence has been found as speedy phenomena; within six months after onset of treatment over 70% isolated S. *aureus* is found resistant to erythromycin. Minimization and mitigation of antibiotics can only be effective way to reduce the severity of such situation. Different technologies for mitigating antibiotics are discussed in the following sub-sections. It is, nonetheless, identified that conventional methods are inadequate to remove antibiotic traces. Therefore, many extensive researches have been done to develop technologies to degrade antibiotics more efficiently and economically. These strategies aim to enhance the removal efficiency by utilizing environmentally benign and methodical procedures with minimized equipment and energy cost.

2. Degradation of antibiotics via physiochemical strategies

Most of the WWTPs and DWTPs are not found suitable for the removal of polar contaminants. Therefore, various economical and in-practical solutions were taken into consideration. A wide range of physicochemical treatment options are employed to tackle the danger caused by antibiotics that includes chemical oxidation, adsorption, membrane techniques (nondestructive processes), coagulation (Homem and Santos, 2011). However, it has been seen that these technologies are plagued with economical and technical barriers, which may include operating cost and production of high amount of damaging sludge. This also should be noted many conventional processes require high energy input and chemicals. The cost and choice of method can be chosen depending upon the impurity and its concentration present in the waste. Different physicochemical strategies are compiled in table 1.

2.1 Oxidative remediation

2.1.1 Chlorination

Chlorine gas or hypochlorite has been conventionally used for water treatment. In addition, WWTPs also deploy chlorination for the oxidation of pollutant residues (Acero et al., 2010). Chlorination is performed to convert pharmaceutical waste to readily less toxic and biodegradable before applying degradation. Standard oxidation potential is highest for hypochlorite (E^0 =1.48 V), followed by chlorine gas (E^0 =1.36 V), and chlorine dioxide (E^0 = 0.95 V) (Sharma, 2008). Chlorine hydrolysis occurs in water and the final products obtained as HOCl, Cl^- and H^+.

Navalon et al (2008a) performed work on β-lactam antibiotics (penicillin, amoxicillin and cefadroxil) using ClO_2. Study revealed that antibiotic degradation increased with increasing ClO_2 concentration as antibiotic react stoichiometrically. Trihalomethane formation was reduced by pretreatment with ClO_2. Another research by Serna-Galvis et al (2017) was performed to treat fluoroquinolones, a highly consumed antibiotic and usually found in waste and natural water. Ciprofloxacin, norfloxacin and levofloxacin were treated with active chlorine. Ciprofloxacin and norfloxacin were removed faster than levofloxacin, as tertiary amine found in levofloxacin were less reactive to active chlorine than secondary amines found in ciprofloxacin and norfloxacin.

Researchers have concluded the beneficial effects of using chlorination method over other conventional techniques like filtration, and coagulation/ flocculation/ sedimentation due to low organic matter load in degrading antibiotics. Besides this, degradation rates were found to increase with increasing pH. But due to the high carcinogenic byproducts produced this procedure was replaced by other advanced oxidation processes.

2.1.2 Advanced oxidation processes

Alternative to oxidation, advanced oxidation processes (AOPs) are being employed to eliminate antibiotics and their byproducts as recalcitrant behavior remains which interfere with removal of antibiotics by traditional biological treatment. This method works on the generation of highly reactive intermediate radicals (OH^-) having redox potential of about 2.8 V, which is quite high than other conventional methods, thus highly efficient in oxidizing most organic compounds (Bautitz and Nogueira, 2007). Free radicals are produced generally by using ozone (O_3), hydrogen peroxide (H_2O_2), or by combining with UV radiation and metallic/semiconductor catalysts. The major advantage of this process is the formation of less refractors byproducts as it has been observed in certain cases that intermediate compounds are more lethal than parent molecule itself. Techniques like ozonation, Fenton, photo-Fenton, and photolysis are discussed in this section.

2.1.2.1 Ozonation

Ozone being a strong oxidant (E^o=2.07 V) is capable of oxidizing major classes of organic materials. Compounds having aromatic bonds, C-C bonds, and nitrogen, sulfur, phosphorous or oxygen atoms can be directly treated with ozone (O_3) and such mechanisms does not fall under the category of AOP (Taylor et al., 2007). While, decomposition of wastes using ozone in water follows different mechanisms like when O_3 reacts with OH^- it forms oxygen and water. The reaction takes place between O_3 and HO_2^- to produce HO_2^* and O_3^{*-}. The HO_3^* then converts into hydroxyl ion and O_2.

A recent research by (Chu et al., 2020) performed on remediation of cephalosporin C, antibiotic of β-lactam class, showed favorable results. Study used ozonation, radiation, and thermal treatment for the degradation and removal efficiency was recorded at 79.9% for ozonation; 85.5% for radiation; 71.9% and 87.3% for thermal process for 4 hours at 60°C and 90°C respectively. Ozonation was found to be more efficient as total content of COD, protein, and polysaccharide was removed greatly. Ozonation is preferred when effluent's composition and flow-rate fluctuates, the high cost or overall mechanism and equipment, and energy consumption are the major disadvantages associated with it. Also, the mass transfer of ozone from gas to liquid phase should be considered as it affects the efficiency and cost of the system.

2.1.2.2 Fenton and photo-Fenton

Among various AOPS, this technique appears to be quite demanding because of the presence of Fenton's reagent, a strong oxidizing agent. Fenton's oxidation occurs in both heterogeneous and homogeneous systems (most utilized) (Gan et al., 2009). It is well known to produce free hydroxyl ions (OH^*) through radical mechanism between Fe^{2+} and H_2O_2.

$$Fe^{2+} + H_2O_2 \longrightarrow Fe^{3+} + OH^- + OH^* \qquad (1)$$

In addition, Fe^{3+} can interact with excess H_2O_2 to form Fe^{2+}. As an alternative, hydroxyl ion interacts with organic pollutants for degradation to occur. Also, in the process of irradiation with visible light, the conversion of ferric ions to ferrous ions was promoted and hence more hydroxyl ions were generated in photo-reduction. The rate of pollutant degradation increases as generation of OH^- ions promotes oxidation of organic compounds (antibiotics) (Liu et al., 2018) . Fe^{3+} ions tend to absorbs light up to 400nm, hence elevating photo-reduction using solar light.

$$FeOH_2 + hv \longrightarrow Fe^{2+} + OH^* \qquad (2)$$

Although, hydroxyl radical formation can occur other than using hydrogen peroxide, for the electro Fenton process, other electrode materials (E) are being utilized for the effective generation of free heterogeneous hydroxyl ions (Liu et al., 2018).

$$E + H_2O \longrightarrow E(OH^*) + H^+ + e^- \qquad (3)$$

According to (Rozas et al., 2010) different variables like pH, Fe^{2+} concentration, and H_2O_2 affect the degradation kinetics of ampicillin. 87 mol/ L; Fe (II) and 400 mol/ L; H_2O_2, Fenton and Photo-Fenton reaction was optimized at 3.5 pH. The complete removal of AMP was achieved at

10 and 3 min for Fenton and photo-Fenton, respectively. For photo-Fenton and Fenton processes the removal percentage of antibiotic was higher than in conventional Fenton method. In addition, UV radiation was found to enhance the degradation rate and complete mineralization was achieved for SMX (42%) and TMP by UV/H_2O_2 process.

2.1.2.3 Photolysis and photocatalysis

The mechanism of photolysis involves the use of light (artificial or natural) for the degradation of effluent to intermediate products, which can be again hydrolyzed to form nontoxic end products (Lofrano et al., 2017). Direct, indirect, and self-sensitization, are the three major pathways of photolysis. In direct photolysis the organic matter can itself absorb light and may breakdown also referred to as photo-ionization, pyrolysis, and heterolysis. Whereas in indirect photolysis production of reactive species such as singlet oxygen ($^1O_2$), hydroxyl radicals (OH^*), and alkyl peroxy (OOR) radicals are used for remediation. Mainly substances present in environment tend to absorb light and produce reactive radicals. Direct and indirect processes can be applied simultaneously, though indirect method was found to be more effective for half-life of the organic matter (Homem and Santos, 2011). In self sensitized photolysis, absorbed light energy causes the formation of reactive oxygen species ($^1O_2$, OH^*), which can ultimately remove ground state molecules to organic substances. The affectivity of the process is greatly influenced by the occurrence and fate of antibiotics in environment, antibiotic structure, pH, toxicity, and degradation products (Shi et al., 2020).

The method is economical as compared with others although it is inefficient to treat environmental membranes contaminated with antibiotics. As the method can only be applied to water sources having photo-sensitive and low COD compounds such as river and drinking water.

2.1.2.4 Electrochemical processes

Oxidation and mild reaction conditions has given strong recognition to the process that has revolutionized the degradation process of antibiotics to the least toxic or non-toxic substances in waste water (Crini and Lichtfouse, 2019). Anodes like TiO_2, graphite, Ti-based alloys, Ru and Ir oxides, boron-doped diamond electrodes (BDD), and dimensionally stable anodes are commonly used. Among these, BDD is the mostly used electrode due to its highest O_2 evolution over weak reaction, hence providing efficient degradation.

$$BDD + H_2O \longrightarrow BDD(^{*}OH)_{ads} + H^{+} + e^{-} \qquad (4)$$

Although, improving anode quality is the matter of research to achieve maximum degradation. Electrochemical oxidation can be performed in two ways: direct and indirect. In direct oxidation, the absorption of pollutants occurs first on anode surface and then remediation occurs by anodic electron exchange reaction. While in latter case the degradation of organic substances take place in bulk liquid by mediating electroactive species (Ag(II), Fe(II), Ce(IV), Mn(III) or strong oxidants such as O_3, H_2O_2, per carbonate, per sulfate, chlorinated, and perphosphate species), which act as intermediates for transferring electrons between organic substances and electrodes (Panizza and Cerisola, 2009). Electrode catalytic activity, current, diffusion rates affects the kinetics of direct electrochemical oxidation, while pH, temperature, and diffusion rates of secondary oxidants are the major controlling parameters for the efficiency of indirect oxidation.

A study performed for the removal of antibiotic levofloxacin (LFX) using electrochemical oxidation process by co-doped PbO_2 electrode. The reaction followed first order kinetic model, and degradation of four aromatic intermediate were identified at pH 3 having the current density of 30 mA cm^{-2}(Xia and Dai, 2018a).

As no external reagents were required, the technique was highly efficient. However, due to the high cost of electrode, the process is still not widely used. Also, to achieve maximum degradation of organic matter, conductivity of the waste water should be high (not a frequent case). Therefore, improving current efficiency and reducing mass transfer resistance is the major subject of condition to tackle.

2.1.3 Adsorption

Adsorption has been identified as an easy method for the separation and removal of organic compounds. Adsorption is identified as the aggregation of molecules from the liquid or gaseous phase to a solid substrate. For the removal of antibiotics, adsorption has not been widely explored even though identified as a potent method for its removal. Adsorption is a surface phenomenon that occurs due to the unbalanced forces between the two phases (Yu et al., 2016). As a result, surface of solid try to attract and retain nearby molecules, hence achieving adsorption. On the basis of forces involved, adsorption can be categorized as physical and chemical. In the former one, weak forces like Van der walls are involved, while chemical adsorption occurs due to electron transfer and chemical bonding (more strong and stable) (Homem and Santos, 2011). Adsorbate and adsorbent properties such as: pore diameter, porosity, and surface area are the major affecting factors of adsorption's efficiency. Other operating parameters such as activated carbon type, activation mechanism, precursors, ionic strength and pH also influences adsorption efficiency.

The most widely used adsorbents for organic matter (antibiotic) removal are: biochar (BC), clay minerals, activated carbons (AC), carbon nanotubes (CNTs), multiwalled carbon nanotubes (MWCNTs), bentonite. According to a case study after examining the adsorption coefficient (K_d) adsorbents trend observed for sulfamethoxazole were BC>MWCNTs>graphite and for tetracycline SWCNTs >graphite >MWCNTs> AC> bentonite respectively as the preferred adsorbents (Ahmed et al., 2015).

Méndez-Díaz et al (2010) analyzed effect of batch adsorption of imidazoles and sulphonamides on activated carbon achieving ~90% removal of both antibiotics. Kim et al (2010) examined the maximal removal percentage of trimethoprim by batch or continuous, techniques. Engineered CNTs have shown greater potential to be used in antibiotic remediation. In a study adsorption on SWCNTs and MWCNTs of lincomycin and sulfamethoxazole was performed, in which SWCNTs were found more potent to adsorbed the pollutant onto their surfaces (Kim et al., 2014). Due to the low cost, high efficiency and no risk of hazardous byproducts, adsorption continues to be employed for separation and removal of antibiotics. Unlike certain processes described earlier, adsorption is also applicable on water containing high level of antibiotics or organic matter with a drawback of only removing and concentrating them.

2.1.4 Membrane processes

Many membrane processes are widely employed in several applications. However, this method only enables the separation and concentration by its transference to the membrane. Reverse osmosis, nano and ultrafiltration, ion exchange, and combined processes have been employed for fulfilling the purpose of antibiotic separation via membrane processes.

Reverse osmosis (RO), nano-filtration, and ultrafiltration (UF), all are the membrane-based separation techniques having different pore size. Reverse osmosis enables the separation of large molecules and ions by applying pressure and a selective semi-permeable membrane. This process is capable of separating dissolved salts in effluent but is limited for separating organic matter. Characteristics of membrane such as its chemical and physical (pore size, mechanical resistance) greatly affect the separation process (Homem and Santos, 2011). Therefore, for the efficient separation, membrane should be stable for long operating processes and resistant to chemical and microbial contaminants.

For the separation of charged entities, ion exchange is most commonly employed. It involves two exchange membranes classified as anion and cation exchangers. Anionic exchangers contain positively charged groups hence capturing negatively charged ions whereas cationic exchangers tend to capture positively charged groups due to negatively charged. Styrenic and acrylic resins based polymeric substances are used as membrane because of their high stability, and greater selectivity. Other materials like bentonite, zeolites or phosphate salts are also used for membrane, the pros with these membranes are comparatively large pore size, bad electrochemical properties, and are very expensive (Nagarale et al., 2006).

For elevating the sensitivity, selectivity, and affectivity for application at industrial level, combined or hybrid processes is one of the potent methods. Therefore, integrated processes have been employed. In the case of degradation, where most of the microorganisms are sensitive towards certain chemicals or mechanism, there is a need for application of combined processes. Hence, AOPs are applied as a pretreatment process, in which organic matter or contaminants are pretreated to convert into less toxic that can easily degrade and remove (Homem and Santos, 2011). Combination of Fenton and RO was applied for the removal of amoxicillin. Liquid-liquid extraction was performed that removed 50% organic matter (TOC), later Fenton oxidation (TOC 38% removed) and lastly reverse osmosis that removed 11% TOC additional, reaching to 99% TOC removal efficiency (Elmolla and Chaudhuri, 2010). Despite being very powerful and potential technique, this method is not applied for the removal of antibiotics due to complexity of the process.

3. Antibiotics removal via biological pathway

Biological degradation has received omnipresent recognition from the scientific community due to many advantages associated in terms of low cost, no toxic by-product generation, simple and ease of operation. As compared to other mechanisms, degradation is identified as a prospective option for antibiotic removal under diverse condition with ability to apply under vast operating conditions and design; degradation can be employed with natural as well as with recombinant microbial cells. Evaluation of the biodegradability of the contaminant is obligatory, in case of biodegradation. According to Organization for Economic Co-operation and Development (OECD), when O_2 used in experimental vessel exceeds 60% of theoretical oxygen in demand for 28 days period, a substance is said to be biodegradable (Phoon et al., 2020).

Biodegradation of antibiotics can be obtained by employing plants, microorganisms (bacteria, fungi, algae), and biocatalysts, each having their certain advantages and disadvantages in antibiotics removal efficiently depending upon temperature, pH, and complexity (structure). According to (Beek et al., 2005) substances showing 20% and 70% biodegradation in Zahn–Wellens test should be considered as "partially biodegradable/ removable" with producing non-toxic stable by-products. Microbial cells can utilize or produce the enzyme in response to the surrounding environment. In biodegradation of antibiotics, ARBs also play crucial role. Corresponding degrading enzymes can be produced by these resistant bacteria, modifying or hydrolyzing antibiotic structure. Algal and fungal strains have also been extensively used for effective removal of antibiotics. Different biodegradation processes with their respective experimental conditions and organisms are mentioned in table 2.

3.1 Bacterial route for antibiotic degradation

In general, remediation involves cleavage of chemical structure with the aid of catalytic enzyme produced by microorganisms. Presence of various gene encoding enzymes (oxidoreductases, ferredoxins, dehydrogenases, oxygenases, glutathione-S transferases, cytochromes, sulfur metabolizing proteins), have been identified in Pseudomonas, showing significance in degradation of organic substances (Dangi et al., 2019). Cephalosporins and fluoroquinolones, highly used veterinary antibiotics, were treated with microbial inoculum isolated from rhizospheric sediments of plant where, ceftiofur (CEF) and enrofloxacin (ENR) were completely degraded using microbial consortia within 21 days (Alexandrino et al., 2017). However, chemical complexity of antibiotics and their derivatives, limit the exploitation of bacteria. Recently, biodegradation of sulfamethoxazole was attained by heterotrophic aerobic and anaerobic bacteria attaining 3% and 2.2% mineralization, respectively (Alvarino et al., 2016), whereas degradation of sulfamethoxazole in fresh water was found to be difficult. In a 28 days study antibiotic component were not degraded while a considerable decrease in acetaminophen and phenylbutazone was found (Baena-Nogueras et al., 2017). Various strains such as: *Microbacterium*, *Stenotrophomonas, Burkholderia*, *Escherichia*, *Labrys, Stenotrophomonas, Ochrobactrum, Acinetobacter, Bacillus, Klebsiella* has been employed for the biodegradation of antibiotics.

3.2 Fungal route for antibiotic degradation

Currently, fungal strains have proven effective in mineralizing and degrading recalcitrant antibiotics due to their enzymatic activity and broad metabolic potential. Fungi release certain extracellular enzyme (ligninolytic enzyme) which causes the degradation of antibiotics by cleaving their parent structures. A study performed on degradation of ofloxacin (OFLOX) and X-ray contrast agent iopromide (IOP) by white-rot-fungus *Trametes versicolor* showed positive results with 60-80% degradation of OFLOX and IOP. Sequential deionization and N-dealkylation were the two principle reactions involved in IOP degradation while, hydroxylation, cleavage of piperazine ring, and oxidation attributed for OFLOX degradation (Gros et al., 2014). Biodegradation of ofloxacin, norfloxacin, and ciprofloxacin was achieved (substitution or decomposition reaction was performed to cleave ring) by using a group of white rot fungi; Panus*tigrinus, Pleurotusostreatus, Irpexlacteus, Dichomitussqualens, Trametes versicolor* in14 days as studied by Liquid chromatography-mass spectrometry for end products. Principal Component Analysis (PCA) of ligninolytic enzymes was performed to elucidate the mechanism of degradation which indicated the presence of manganese peroxidase for remediation (Čvančarová et al., 2014).

3.3 Algal route for antibiotic degradation

Treatment of contaminant for environmental protection has attracted researcher's attention in recent years. Many physical and chemical processes employed such as oxidation, usually achieve low mineralization of chemical entities and may produce more toxic products than the parent compound itself. Moreover, biological treatment (activated sludge processes) was found to be less efficient with increasing concentration (Homem and Santos, 2011). Therefore, a safer, greener, and efficient method was introduced in form of green algae. Due to the presence of higher biomass and resistance to pollutant, algae have identified as a potent alternative for degradation of antibiotics. Green algae (a non-target organism) have shown high removal efficiency and tolerance towards antibiotics than other algal species.

In a recent study, microalga, *Scenedesmus obliquus* was investigated for imposing toxicological effect by fluoroquinolone and levofloxacin (LEV) antibioticin aquatic reservoirs. In the absence of NaCl, removal of LEV was less, which significantly increased from 4.5 to 93.34%; with increasing salinity (0 to 171mM NaCl), Kinetic studies in presence of NaCl represented decrease in half-life from 272 to 5 days and increase in removal rate constant (k) from 0.005 to 0.289 /day. *S. obliquus* found to perform metabolic reaction including demethylation, dihydroxylation, decarboxylation, ring cleavage, and side chain breakdown (Xiong et al., 2017). In another investigation, degradation of sulfamethazine (SM_2) was performed by *Chlorella pyrenoidosa*. For this purpose, growth inhibition effect on *Chlorella pyrenoidosa* of SM_2 (2, 4, and 8 mg/L) for two weeks was monitored. The peak accumulated concentration of SM_2 in *C. pyrenoidosa* on day 13 were 0.225, 0.325, and 0.596 ng per mg, showing potential to degrade SM_2 (Sun et al., 2017). Rapid adsorption, slow cell wall transmission, and degradation are the three major steps of algal removal.

3.4 Phytoremediation of antibiotic

The growing demand for cost effective and on-site treatment of antibiotics has fastened the technologies to find upgraded and efficient forms of degradation. Recently, a promising technology, phytoremediation has been extensively studied in laboratory and pilot scale (Hoang et al., 2012). Phytoremediation is applied as an alternative of AOPs. This process involves the use of vegetation for modification, removal or degradation of antibiotics. Tetracycline and oxytetracycline (OTC) has been removed using *Pistia stratiotes* (water lettuce), *Helianthus annuus* (sunflower), and *Myriophyllum aquaticum* (parrot feather). Highest removal was by water lettuce, intermediate by sunflower while parrot feather gave least, which was dependent on growth period of each plant (Gujarathi et al., 2005).

In a recent study, performed to study degradation of tetracycline (TC), Vetiver grass was used as a phytoremediation agent. The vetiver saplings were germinated for over 60 days in tetracycline contaminated hydroponic system. Within 40 days complete removal of TC occurred, and was found to be present in tissues of root and shoot, confirming their uptake and root to shoot translocation (Datta et al., 2013). For instance, *Myriophyllum aquaticum* (Vell.) Verdc., *Zea mays* L. *(corn), Phragmites australis* (Cav.) Trin. Ex Steud., *Lupinusalbus* L., *Allium cepa* L. (onion), *Pistia stratiotes* L., *Brassica oleracea L. Capitata* group (cabbage), and *Hordeum vulgare* L. have been used for successful phytoremediation of organic matter.

3.5 Mechanism of degradation of antibiotics

As described, degradation initiates with breakdown of chemical bonds within the parent molecule of antibiotic. However, the degradation route majorly depends on the chemical nature of antibiotic and on factors like duration, type of treatment, pH, temperature, and mechanism employed etc. UV/Vis spectroscopy, high performance liquid chromatography, mass spectroscopyis used for determination of possible removal or degradation pathway.

3.5.1 β-lactam antibiotics

β-lactam antibiotics are bactericidal and works by inhibiting the bacterial cell wall synthesis. Due to their low toxicity and broad-spectrum property, these are used all over the world prevalently in both human and animal treatments. A detailed database is present in National Center for Biotechnology Information (NCBI) on various β-lactam antibiotics (Naas et al., 2017). Due to the overuse of antibiotics, resistance and pollution has been created in water bodies mainly via two mechanisms: (i) hydrolysis of β-lactam ring (ii) by mutated penicillin binding protein. In a research performed on amoxicillin, biotic and abiotic processes were included to achieve the mineralization. In process, drug was first hydrolyzed to amoxicilloic acid (abiotic) and then by microbial activity in activated sludge (Längin et al., 2009).

3.5.2 Tetracyclines

Tetracyclines is a bacteriostatic antibiotic and works by preventing the transfer of aminoacyl-tRNA to ribosomal acceptor site, thus inhibiting protein synthesis (Ian and Marilyn, 2001). Many natural, synthetic and semi-synthetic tetracyclines like oxytetracycline, chlortetracycline, glycylcyclines (3^{rd} generation tetracycline) are have been extensively used in human and veterinary medicines (Ian and Marilyn, 2001; Reis et al., 2020a). Degradation of tetracycline has been majorly performed by fungal strains (*Trichosporon mycotoxinivorans* XPY-10, *Trichoderma deliquescens* RA114, *Trichoderma harzianum* RA115, *Peniciliumcrustosum* RA118) and minorly by bacterial strains (*Stenotrophomonas Maltophilia, Sphingobacterium* sp.

PM2-P1-29). Certain enzymes like Tet37, ligninolytic enzymes such as manganese peroxidase and lignin and genes like TetX has been identified in degradation of tetracycline antibiotics (Reis et al., 2020a).

3.5.3 Sulfonamides

Sulfonamides antibiotics are active against most gram-positive, gram-negative, and protozoa (Larcher and Yargeau, 2012). Due to its extensive use it has been found in municipal wastewater, surface water, soils, sediments, and groundwater. Sulfonamides are bacteriostatic in nature and works by inhibiting dihydropteroate synthetase (DHPS), thus blocks the synthesis of folic acid. Photolysis applied for degradation of sulfonamides was found to produce toxic end-products, demanding more potent remediation strategies. Various biotransformation processes like the very first-time sulfonamide biodegradation was achieved in activated sludge. In the presence or absence of carbon and nitrogen sources, biodegradation can be performed. Though it has been seen that complete mineralization with no toxic effluent production was achieved in the presence of nitrogen with the conversion into 3-amino-5-methylisoxazole (3A5MI) (Reis et al., 2020b).

3.5.4Aminoglycosides

The first aminoglycoside to be discovered was streptomycin, which was derived from *Streptomyces griseus.* Within 30S ribosomal unit, streptomycin binds to 16S rRNA and S12 protein in an irreversible manner. It inhibits the formation of initiation complex between bacterial ribosome and mRNA, hence interferes with the protein synthesis. Moreover, streptomycin causes translational frame shift by inducing misreading of mRNA template. Therefore, results in premature termination of the process leading to bacterial cell death. Thus, act as bactericidal antibiotic. Degradation by abiotic method demonstrated at higher temperature and at low pH lead to increased degradation. Studies with kanamycin and gentamycin in high acidic environment (6N HCl), that releases 2-deoxystreptamine causing complete removal of antimicrobial activity of the both antibiotics. Photolysis method was found to be insignificant as aminoglycosides are not able to absorb solar light (Li et al., 2016).

3.5.5 Quinolones

Quinolones works by interfering with the bacterial replication machinery, transcription, DNA topoisomerase IV and II (DNA gyrase). Flumequine, nalidixic acid, and oxolinic acid are the first-generation quinolones and now are rarely used. Fluroquinolone, a second-generation quinolone is used frequently for human therapy. Bacterial strain *Labrys portucalensis* F11 (Carvalho et al., 2016) and brow-rot fungus *Gloeophyllum striatum* (Parshikov and Sutherland, 2012) has been studied in the degradation of fourth-generation quinolones. However, the complex reactions involved in the mineralization of quinolones is limited as nucleus is difficult to cleave. Though, two strains *Curvularia lunata* and *G. striatum* were reported for significant removal of quinolones, 30% and 20%, respectively. In which *Curvularia lunata* has converted 14C-danofloxacin into CO_2 and *G. striatum* mineralizing ^{14}C-ciprofloxacin or ^{14}C-enrofloxacin.

4. Biological degradation of antibiotics in bioreactor

Bioreactor or fermentor is a term given to a vessel which is employed for biological reactions to take place. Bioreactor provides specific conditions such as: biological system, nutrients, pH, temperature, aeration, and agitation, that is required for a smooth and efficient biological reaction. A bioreactor can include microorganisms, enzymes, tissues, plant, and animal cells. For a productive biological reaction, it is necessary to optimize the environmental conditions as well as understanding of the system (cell growth, genetic manipulation, and metabolism). Moreover, temperature, pH, mixing, dissolved oxygen concentration, and nutrient supplementation considered as variable factors, also needs to be optimized (Zhong, 2010). Since it has been seen that all biological processes are not capable for complete mineralization of antibiotics, these reactors might be used in combination with chemical and physical processes.

4.1 Activated sludge bioreactor

In activated sludge bioreactor, pollutants are aerated to promote bacterial growths that gradually consume toxic organic substances. The whole system is composed of two components: an aerator and a clarifier. Aerator aids in growth of cell for substrate consumption while clarifier helps in removal of treated effluent. A work performed using activated sludge bioreactor under redox conditions (aerobic, nitrifying, and anoxic) demonstrated biodegradation of 6 quinolones antibiotics (Dorival-García et al., 2013). Working and design of activated sludge bioreactor is represented in figure 1(a). Due to low cost of equipment and large amount of effluent, activated sludge bioreactor is commonly used with certain drawbacks like continuous and regulated need of oxygen for optimum aeration.

4.2 Expanded granular sludge bioreactor (EGSB)

Expanded granular sludge bioreactor is the combination of fluidized bed (FBR) and up flow anaerobic sludge blanket (UASB) bioreactor. EGSB is a popular due to its robust economic processing and capacity to retain increasing influent load and cell retention, thus achieving efficient removal (Cruz-Salomón et al., 2019). In order to investigate the EGSB removal efficiency, a setup was designed to remove amoxicillin (AMX) for 241 days where 80% and 85% removal was achieved in terms of AMX and COD, respectively (Meng et al., 2015). It is important to maintain temperature, pH throughout the process as kinetics of the system is highly dependent on the effluent characteristics and on the performance of such bioreactors

4.3 Membrane reactor (MBRs)

Membrane bioreactor performs biodegradation with a separation using a membrane and is mainly composed of a bioreactor tank and a membrane chamber. A research investigated on anoxic/ oxic-membrane bioreactor showed high level antibiotic degradation, due to strong hydrolysis efficient removal of penicillin and chlorotetracycline at 97.15% and 96.10% was achieved, respectively ; 90.07% of sulfamethoxazole was degraded and 63.87% of norfloxacin was removed mainly via sorption(Wen et al., 2018). In another recent study, aerobic submerged membrane bioreactor was used to achieve degradation of sulfadiazine (SDZ) and SMZ91% and 88%, respectively (Yu et al., 2018). MBRs have the advantage over conventional reactors as it provides high sludge retention time (SRT). MBR have shown better removal efficiency in comparison to the sludge bioreactor as membrane can filter adsorbed pollutants from suspended solids. However, due to the presence of membrane, fouling occurs that has to be regulated frequently (de Cazes et al., 2014).

4.4 Enzymatic membrane reactors (EMRs)

Recently, a new bioreactor containing membrane and biocatalyst have been explored. These reactors can be configured in two ways: (i) by coupling filtration unit with enzymatic reactor and membrane acting as selective barrier (ii) selective membrane and enzymes in single cabinet and membrane used as the support for immobilized enzyme. The latter being suitable for the removal of reluctant antibiotics and has been employed in removal of pollutants including dyes, bisphenol A, carbamazepine, atrazine, diclofenac, and sulfamethoxazole (de Cazes et al., 2014). Tetracycline degradation using *Trametes versicolor* in EMR using immobilized and free laccase shows efficiencies of 56% and 30% respectively (Abejón et al., 2015). Though, enzymatic degradation of antibiotics has come up as a potent strategy, it is important to understand the kinetics and membrane properties to establish the capex and opex competitiveness with other processes.

4.5 Combined/ hybrid processes

Other potential and efficient bioreactors *viz* anaerobic immobilized biomass, stirred tank, and membrane biofilm reactor are also available for the biodegradation of organic matter present in water bodies. Though these strategies provide efficient degradation of antibiotics, certain more potent reactor designs are required to provide complete removal. As the single treatment methods are found inefficient to remove recalcitrant antibiotics. In such cases, it is recommended to use combined/ sequential or hybrid treatment options to meet required level of degradation. In sequential processes two or more different processes are combined together in a sequential manner. In contrast, hybrid processes are considered as more cost effective and less time consuming than combined processes as only one chamber is required to perform multiple processes.

Sequential batch reactors (SBR) are used for wastewater treatment involving four basic steps filling, aeration, settling and decantation. That employs treatment of unwanted substances in the chamber and then discharge off the treated effluent. In an investigation removal of cefradine (60 mg/L) was performed using green algae in SBR, with ~ 76.02% removal in 96 hours (Chen et al., 2015). Likewise, a system for removal of sulfonamides, tetracycline, and tiamulin by ozone with sequential batch reactor has been demonstrated. The degradation efficiency was significantly increased by ozone treatment in SBR.

5. Use of biocatalyst for degradation of antibiotics

Enzymes are the biological catalysts, isolated from the biological sources (plants, microorganisms). They catalyze the reaction by converting the substrate into desired products. Moreover, enzymes are specific and can discriminate between slightly different substrate. Due to the several disadvantages of previously discussed strategies, environmental engineers utilized enzymes as robust and efficient alternative for degradation of antibiotics. Exploitation of biocatalyst-based methods, has unveil new horizons to remove recalcitrant contaminants.

Recently enzymatic degradation are successfully employed for oil degradation in fields/ sea, biosensors, biofuel cells, decolorization of textile dyes etc. (Vikrant et al., 2018).

However, instead of their novel features, enzymes are not being deployed at industrial scale as enzymes in free forms are highly unstable, difficult to separate, inconvenient to handle, non-recyclable, intolerant to unfavorable conditions, and also posed high processing cost (Bilal et al., 2019c) . Therefore, it is desirable to create encapsulate or immobilized enzymes with high catalytic property for achieving high degradation efficiency. Different methods of enzyme immobilization such as cross-linking, organic/inorganic polymer binding, and encapsulation have been utilized frequently. Also, magnetic nanoparticle, stevensite or electrospun poly (lactic-co-glycolic acid) nanofiber, enzyme-embedded membrane, and magnetic cross-linked enzyme aggregate (M-CLEAs) use of biochar etc. have been extensively used to improve stability, reusability, durability and catalytic efficiency of enzymes (Bilal et al., 2019d).

5.1 Mechanism of enzymatic degradation

Currently crude/purified immobilized enzymes have been investigated for the transformation of pharmaceutical contaminants (Alneyadi et al., 2018). Extraction of enzymes from the highly efficient species such as white-rot fungi as an alternative treatment to conventional WWTPs. Various enzymes including lignin peroxidase (LiP), laccase, manganese peroxidase (MnP), and versatile peroxidase (VP) have been explored for the degradation of antibiotics. Peroxidase enzymes can be characterized as heme-containing or non-heme containing, former (Fe^{3+} form) reacts with H_2O_2 to generate cationic compound-I (Fe^{4+} form). This active form of enzyme reacts with organic substances to generate radicals and converted to another form (Fe^{4+}) of enzyme, compound-II. This compound-II can further react with organic pollutant converting it into its native form (Fe^{3+}). The resultant contaminant radicals react with each-other to degrade in reaction.

$$\text{Heme peroxidases} + H_2O_2 \longrightarrow \text{compound I} + H_2O \quad (5)$$

$$\text{Compound I} + \text{organic pollutant (RH)} \longrightarrow \text{compound II} + R^o \quad (6)$$

$$\text{Compound II} + RH \longrightarrow R^o + \text{heme peroxidase} \quad (7)$$

5.2 *Removal efficiency of enzymatic degradation*

In a study, chloroperoxidase (CPO) entrapped in nanocomposite of dendritic silica particles (DSP) (CPO@DSP) were investigated for the removal of levofloxacin and rifaximin (Song et al., 2019). Nanocomposite form of enzyme could retain 90.74% activity at 80^{o}C, after 3 hours of processing while free form of enzyme could retain only 14.28%. In the same work, a construct made with amyloid-like protein nanofilm with phase transition lysozyme (PTL) was coated with CPO@DSP surface and then PTL was cross-linked to another CPO surface constructing CPO@PTL-CPO@DSP. The construct removed 88% of both antibiotics within 30 minutes. In another research, synthetically immobilized laccase enzyme on yeast cell, referred as surface display laccase (SDL) was used for the removal of sulfamethoxazole and bisphenol A (Chen et al., 2016). After 25 days storage 90% initial enzyme activity was retained by immobilizing while free laccase enzyme activity was depleted to 60%. Mechanism of immobilized enzyme and laccase enzyme has been represented in figure 1(b).

Working with biocatalyst requires consideration towards physico-chemical parameters such as optimum pH, temperature, initial substrate concentration, and enzyme concentration. As enzymes are pH and temperature sensitive, these parameters impose intense effect on rate of reaction and removal of antibiotic from the system. In a study, performed on removal of erythromycin by erythromycin esterase showed maximal degradation at optimum conditions of pH 6 and temperature 45^{o}C (Cazes et al., 2016). The efficiency of enzyme immobilization on membrane depends upon porosity, properties, and material characteristics; properties of enzyme, and method of immobilization (entrapment, covalent or non-covalent bonding, gelification).

Immobilized enzymes are also beneficial and a better alternative to whole-cell containing systems. As it is easy to separate immobilized catalyst by simply pouring them into other solution, while for whole cell separation centrifugation is mandatory, causing difficulty in recovery for reutilization. Use of immobilized cells makes downstream processing uncomplicated and thus more cost-effective (Sharma and Leblanc, 2017).

6. Microbial fuel cells (MFCs) for removal of antibiotics

Challenges encountered (cost and energy intensive) in conventional bio-based remediation strategies such as anaerobic or aerobic digestion and sludge processes lead to probe into various other cost and energy efficient methods. It is worth noticing that high concentration of suspended and dissolved organic substances in effluent source store a potent amount of energy ranging from 4.92 to 7.97 kWh $kgCOD^{-1}$ (Dong et al., 2015). Hence, MFC emerges as potent environmental technologies that integrate electrochemical and microbial processes and can convert potential energy (80%) to bioenergy by metabolizing organic load present in effluent. In comparison, MFC requires less energy, generates electricity and utilized for removal of variety of pollutants including heavy metals, vaporization of nitrogen, sulfate reduction, inorganic non-metals, antibiotics, hydrocarbons, and decolorization of dye wastewater (Li et al., 2020).

6.1 Configuration of MFCs

A basic design of the process includes a separating membrane (separator), an electrode, and their arrangement (Bakonyi et al., 2018). Depending upon the design and layout it can be categorized as single and double chamber MFCs. Biocatalyst at cathode and anode plays a significant role in imparting electricity and efficiency of MFCs (S. S. Kumar et al., 2019).

In single chamber MFCs, cathode is found in contact with liquid and the other side is exposed to air directly. A microfiltration membrane is found on the inner surface of cathode. Oxygen present in air can diffuse via cathode and perform oxygen reduction. This process does not require liquid aeration which makes it low energy and cost effective. In a study, 88% of neomycin sulfate was removed using single chamber MFCs and was monitored by LC-MS/MS, for COD, and total carbohydrate removal (Catal et al., 2018).

In dual chamber MFCs, the whole setup is divided into two compartments, cathode and anode arm. Both chambers are separated by an ion selective membrane such as proton exchange membrane (PEM) maintained at pH 7. Pollutants get oxidized in anodic chamber by microbial strains. The generated protons then pass through PEM and e^- migrate through external circulatory system at cathode with electron acceptor (Zhang et al., 2016).

6.2 Microbial strains used in MFCs

In nature naturally found microbes are electrochemically inactive, to be employed in MFCs; high opex mediators (methyl viologen, natural red, thionine, methyl blue) are used to facilitate the transfer of electrons. Hence, electrochemical active species like *Geobacter, shewanella, Pseudomonas, Cupriavidus, Rhodoferax, Propionibacterium,* and *Lactococcus sp.* are preferred (Rahimnejad et al., 2015). However, there are certain other factors responsible for generation of bioelectricity. Electrode size, pH, resistance of the system, spacing between electrodes, conductivity, and architecture of the system etc. are the major parameters to be considered. Moreover, degradation of antibiotics and other pollutants in MFCs also depend upon the species of microorganism, composition, and its acclimatization (Bakonyi et al., 2018).

6.3 Mechanism of antibiotic removal via MFCs

MFC is composed of an aerobic cathode and anaerobic anode that may or may not be separated by ion selective electrode (adhered on cathode using hydrogels or ion exchange resins). This ion selective electrode regulate the flow of protons from anode to cathode (Daud et al., 2020). Both mounted electrodes are joined together via an outer circuit. Cathode terminal is fed with substrate degraded by bacteria and electrons generated are transferred via anode to cathode through outer circuit responsible for electricity generation. Simultaneously, generated protons react with electronic acceptor to form water.

$$CH_3COO^- + H_2O \longrightarrow 2CO_2 + 2H^+ + 8e^- \qquad (8) \text{ anodic reaction}$$

$$O_2 + 4e^- + 4H^+ \longrightarrow 2H_2O \qquad (9) \text{ cathodic reaction}$$

$$\tfrac{1}{2}\, C_6H_{12}O_6 + H_2O + 3O_2 \longrightarrow 3CO_2 + 4H_2O \qquad (10) \text{ overall reaction}$$

Antibiotic degradation in MFC can generally be understand by: (i) antibiotic act as carbon source in bio-anode; (ii) degradation of antibiotic occurs by microorganisms on cathode; (iii) free radicals degrades (produced by cathode-modified materials) antibiotics (Yan et al., 2019). A study performed on removal of sulfamethoxazole using MFC showed 85.1% degradation at 60 hours. HPLC was utilized to identify the end-products (less harmful alcohols and methane). *Pseudomonas, Alcaligenes,* and *Achromobacter* were involved in SMX degradation while *Geobacteria sp.* and *Shewanella sp.* in power generation. In addition, to antibiotic removal it has been observed that low copy number of integrons and ARGs were produced as compared to the conventional WWTPs (Xue et al., 2019).

Due to the high opex, use of MFCs is limited which can overcome by efficient designing of electrode; recently, carbon and metal-based electrodes have been explored (Rahimnejad et al., 2015). Advanced electrode like biocathode eliminates the use of noble metal-based electrodes and electron mediators. MFC is not just restricted to wastewater treatment; a new MFC process BEAMRs also being utilized for the generation of bio-hydrogen from biodegradable matter. Though, antibiotics can be utilized in MFCs as a carbon source for power generation. It has been observed that low power could encourage transfer of ARGs. Therefore, ambient power generation and antibiotic concentration is critical to achieve efficient operation.

7. Role of biochar for antibiotic removal

Use of biochar has been recently introduced in elimination of pharmaceutical substances. It is an excellent carbon-rich sorbent derived from biomass for the removal of hydrophobic organic compounds. Due to its aromaticity, surface adsorption properties, functionality, and cost effectiveness, biochar based removal techniques has received enormous investigation (Vithanage et al., 2016). Porous structure, surface functional groups, and area makes it a potent adsorbent for removal of contaminant (Zhu et al., 2018). Although, the biochar surface is similar to activated carbon but it is more energy efficient, cost effective, and renewable resource. At present, biochar is extensively employed for removal of pharmaceutical products, polychlorinated biphenyls (PCBs), industrial dyes, oils, polycyclic aromatic hydrocarbons (PAHs), pesticides, volatile organic compounds (VOCs), persistent organic pollutants (POPs), and inorganic (nitrate, phosphate, and sulfide) and metalloids.

7.1 Sources and production of biochar

Biochar is mainly produced from lignocellulosic biomass including wood, crop residues, solid waste, and animal dung. Biomass wastes are considered to be most appropriate materials for biochar production that may include food and forestry waste, animal manures, crops, bagasse, and sludge. Composition of biochar largely depends upon the temperature and duration of pyrolysis and type of raw-material (Tag et al., 2016). Biochar is produced by the thermochemical reactions such as hydrothermal carbonization, gasification, torrefaction, and pyrolysis using carbonaceous raw materials under oxygen-limiting environment at high temperature (300-900^{o}C).

7.2 Mechanism of antibiotic removal via biochar

Biochar reduces carbon emission from degradation or combustion by sequestering carbon. It has been reported that biochar has strong adsorption ability for different organic and inorganic substances. Almost all carbon-based adsorbents employ same mechanism that can be electrostatic, physical, or chemical adsorption. Majorly, electrostatic, van der waals, hydrogen bonding, Lewis acid-base, π-π interaction, and π-π EDA interactions are found between adsorbent and adsorbate. Moreover, presence of functional groups such as -COOH, -NH_2, -OH also affect the efficiency of adsorption (Xiang et al., 2019). In another investigation, magnetic biochar (M-BC), produced from *Astragalus membranaceus* (herbal medicine waste) was used for removal of ciprofloxacin. Langmuir and pseudo-second order kinetic model were better fitted and appropriately described the adsorption process. Efficient removal of ciprofloxacin was achieved at pH 6 (Kong et al., 2017). Moreover, various biochar composites such as magnetic $MnFe_2O_4$/bio-char composite for tetracycline (Lai et al., 2019), biochar-supported Fe composite (Fe/C)for sulfapyridine (SPY) and oxytetracycline (Z. Li et al., 2020), immobilized laccase on Polyacrylonitrile-biochar composite nanofibrous membrane for chlortetracycline (Taheran et al., 2017) degradation have also been employed to exploit the advantages of biochar over conventional strategies.

7.3 Advantages and limitations of biochar-based removal

For removal of organic pollutants, researchers have exploited variety of adsorption-based removal from conventional activated carbon to advanced carbon-based biochar materials. Due to its high porosity and changeable surface chemistry, biochar is under investigation for various environmental applications. Moreover, its great adsorption potential and hydrophobicity, that makes it applicable in every possible removal application (Xiang et al., 2019). However, instead of being cost effective and energy efficient, the use of biochar in batch equilibrium is limited. In order to modify adsorption sites and increase surface area, biochars are treated with certain metals (Fe^{2+}, Zn^{2+}, Fe_3O_4) that can cause the emergence of other environmental pollutants. Moreover, the disposal of carbon-based adsorbents is also a matter of consideration.

8. Future scope

In recent years, the fate and occurrence of SOCs like antibiotics has received special attention from biotechnologist for its removal from environmental matrices. Reluctant antibiotics and their by-products are strenuous to degrade and hence create aquatic and terrestrial pollution. Most conventional methods, WWTPs are replaced by advanced oxidation processes like ozonation and Fenton/ Photo-Fenton oxidation. Although, these strategies provide maximal degradation but generation of toxic by-products makes them undesirable. Combination of these methodologies with biological resources may be considered in order to enhance environmental benignity.

Upon strategic evaluation, microbial-based removal of antibiotics still needs considerations as it depends upon the acclimatization of microbial consortia and growth parameters. It is critical to analyze and identify the optimum conditions (pH, temperature), homogenization, dissolved oxygen concentration, oxygen transfer, operational stability and suitability to scale up the process and implementing at industrial level. Effectiveness of the biological system can be enhanced by using a mixed consortium of microbes or by employing combined processes. In addition, recombinant DNA technology can be considered to modify the genome of microbial strain by inserting antibiotic resistant genes, in order to provide exceptional stability and adaptability for upgraded biodegradation for highly reluctant antibiotics.

Analysis of alternatives revealed use of immobilized biocatalysts that can amplify the degradation process surpassing the time required from inoculation to achieve optimum growth and are more specific, reusable hence impose low operational cost. However, instability of biocatalyst and sensitivity to the continuously altered environment limit their applicability which enforces their further investigation. To end this, use of MFCs and biochar-based reactors has provided highly profitable alternative for effective removal of recalcitrant antibiotics. Low cost, low energy requirement and environment friendly nature makes them preferable over other previously discussed methods. Owing to its small size, the operational time of the MFCs process is short that interfere with removal efficiency. Also, it is important to understand the metabolic behavior of electrically-active microbes to attain sustainable removal. In addition, understanding of energy harvesting system is highly recommended for storage and amplification of low intensity energy generated. Similarly, biochar-based system works well for removal of antibiotics by adsorbing them onto their surface. However, use of metals in order to modify structure and surface area, alter their biological properties and leads to the formation of maligned by-products. An estimated factor based on cost of raw material is another important aspect for running operations economically. Hence, development of innovative scientific advancements necessitates bioengineers and environmentalists to implicate combined/ hybrid processes to enhance remediation of synthetic organic pollutants without hampering environment and ecology. Combined endeavors would help in rationalizing designs for removal and ultimate degradation of SOCs such that the current scenario of ARGs and ARBs can be overcome and sustainable healthcare and surroundings could be created.

9. Conclusion

Mitigation of antibiotics and related toxic by-products are necessary to maintain viability of natural habitat. Various potent methodologies including physico-chemical and biological have been utilized for waste water treatment however; their efficacy is limited due to high energy consumption, cost, difficulty in scale-up, generation of unwanted hazardous sludge, and requirement of large amount of raw materials. Ultimately, various methods to remove antibiotics from waste water have been reported and author suggests that the future research should be based on hybrid/ combined technologies for instance; combination of Advanced Oxidation Processes and Membrane separation should be implemented.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

The authors are highly thankful to Director, CSIR-Indian Institute of Toxicology Research (IITR), Lucknow (India) for his valuable encouragement and intra institutional financial support for this research work. This is CSIR-IITR manuscript number is 3684.

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Declaration of interests

☑The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

The authors declare that they have no conflict of interest.

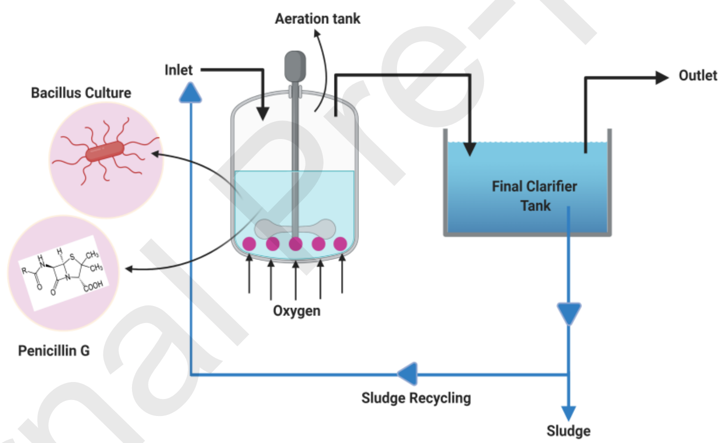
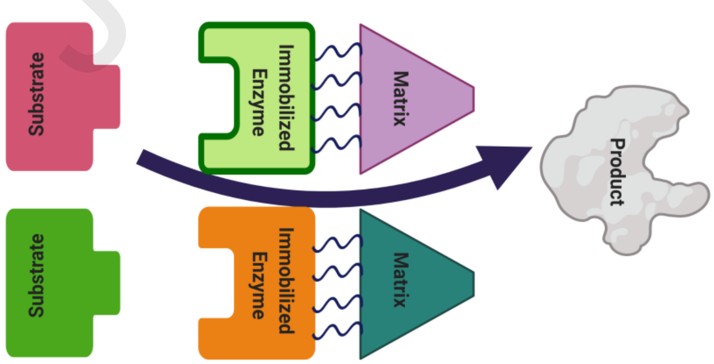
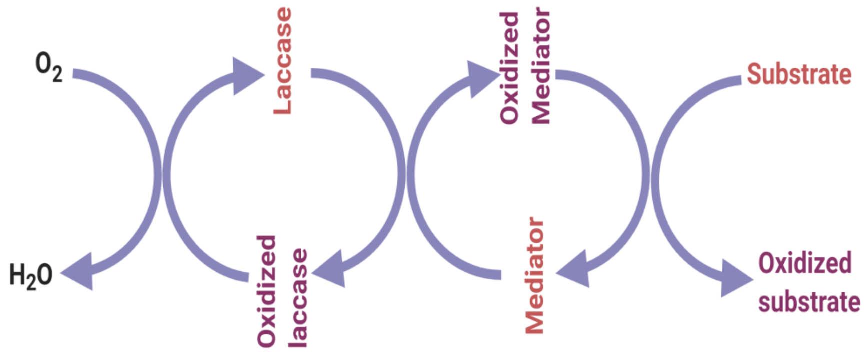


Figure 1 (a): Activated sludge bioreactor. Sludge recycling can be performed to achieve optimum removal of pollutants





(A) (B)

Figure 1 (b): A) Mechanism of immobilized enzyme-based removal of antibiotics B) Activity of laccase enzyme for degradation of phenolic and pharmaceutical compounds in presence of redox mediator (Adapted and reproduced from de Cazes et al., 2014).

Table 1: Physiochemical treatment strategies, operating conditions, and lab-scale output

rder	Antibiotic	Treatment	Operating Condition	Analytical Method	Results	Reference
	Ciprofloxacin; Norfloxacin; Levofloxacin	Electrochemical oxidation by electrogenerated active chlorine	NaCl (0.01 to 0.05 mol/L); Anode= Ti/IrO_2	UHPLC	100% CIP and NOR while 75% LEV was degraded.	(Feng et al 2018)
	Ciprofloxacin	Fenton's reagent (oxidative degradation)	pH=3.0, temperature=30°C; H_2O_2 to Fe^{2+} molar ratio ranging from 5 – 50	TOC, HPLC	At $[H_2O_2]$:$[Fe^{2+}]$ molar ratio of 10, CIP 70% and TOC 55% were removed in 60 minutes.	(Gupta anc Garg, 2018)
	Levofloxacin	Electrochemical degradation	PbO_2 electrode, La- PbO_2 electrode, Y- PbO_2 electrode and La-Y- PbO_2 electrode; current density 30 mA cm^{-2}	TOC, HPLC	With La-Y- PbO_2 95.4%, with PbO_2 83.3%, with La- PbO_2 90.9%, with Y- PbO_2 85.6%, LFX was degraded in 2.5 hour of electrolysis.	(Xia and Dai, 2018)

Cephalosporin C	Ionizing radiation, Ozonation, Thermal treatment	Gamma radiation temp= 20 ± 2°C. Ozone flow rate= 0.6 L/min. Thermal treatment temp= 90±1°C and 60±1°C.	HPLC, COD	94.2% degradation rate for CEP-C by gamma radiation (150 kGy), 79.9% by ozonation (10.4 gO_3/L), and 71.9% (60°C) and 87.3% (90°C) by thermal treatment for 4 hours	(Chu et al. 2020)
Amoxicillin; Ampicillin; Cloxacillin	Fenton	pH=2-4; Molar ratio of H_2O_2/COD=1.0-3.5 and Molar Ratio H_2O_2/Fe^{2+} = 2-150	HPLC-DAD, COD, BOD5, TOC, DOC	Complete degradation in 2 min at pH=3; 81.4% COD and 54.3% DOC removed in 10 min.	(Jain et al. 2018)
Penicillin G	Ozonation (cerium-loaded natural zeolite)	Ce loading mass = 4%, CZ dosage = 2 g/L, ozone dosage = 6 mg/min and the initial solution pH = 4.5	TOC, HPLC, UHPLC, MS	Removal efficiency of penicillin G reached 99.5% after 15 min and TOC removal 21.2% after 2 hours.	(J. Zhang e al., 2020)
Ciprofloxacin	Advanced oxidation processes: UV, H_2O_2, UV/H_2O_2, modified Fenton (nZVI/H_2O_2) and modified photo-Fenton	Reaction time=120 min and Different H2O2 doses: 1 to 100 mmol/L while modified photo-Fenton for 45 min and H2O2 dose of 100 mmol/L with nZVI dose of 5 mmol/L	UHPLC, MS, TOC	60% CIP and 4% TOC removed by direct photolysis in 120 min, H_2O_2 removed 40% CIP in 100 min, different combination of UV, H_2O_2, and nZVI/H_2O_2 removed 100% CIP.	(Mondal e al., 2018)
Enrofloxacin	magnetic montmorillonite (Fe_3O_4/MMT) composite with the function of adsorption	Fe_3O_4/MMT (100 mg/L) was used at 25°C. initial concentration of ENR was 30 mg/L (pH=6.85)	TOC, HPLC, LCMS-IT-TOF	degradation efficiency was 90% in 60 min for ENR.	(Peng et al 2019)
Chloramphenicol	Semiconductor photocatalysis	UV at 320-400 nm; pH=5; 0-4 g/L catalyst; T=3-57°C	Absorbance at 276.5 nm; TOC, antimicrobial activity to *E. coli*	With increasing H_2O_2 drug and catalyst concentration loading rate of Pseudo 1st order kinetic increased; Photo-degradation affected temperature. ZnO and TiO2 catalysts found to be most efficient. With 70% of mineralization complete degradation of antibiotic achieved after 90 min	(Pi et al., 2018)

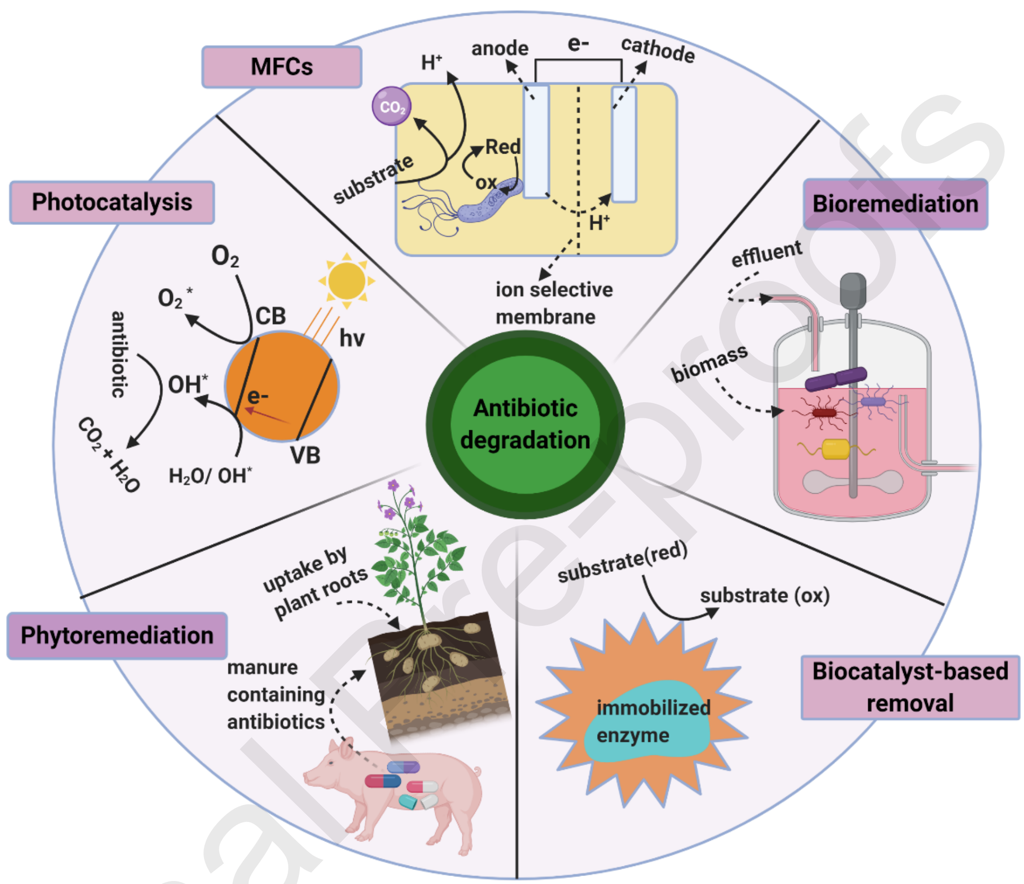
0	Chloramphenicol	Chlorination (chlorine and chlorine dioxide)	Cl_2 concentration 1.0 – 4.0 mg/L and ClO_2 concentration either 1.0 or 2.0 mg/L. Temperature = 25°C	GC-EI-MS, UV-Vis-Spectrophotometer	Removal achieved was 76% with chlorine and 52% with ClO_2. Three disinfection byproducts (chloroform, bromodichloromethane, and dibromochloromethane) were observed at 7.7, 0.76, and 0.39 µg/L after 160 min.	(Li et al., 2019)
1	Ampicillin	Cu (II) oxidation	pH= 5.0, 7.0, and 9.0 and oxygen concentration= 0.2 and 6.2 mg/L	LC-quadrupole-TOF-MS	AMP degradation rate was increased in 6.2 mg/L O_2 solution from 21.8% to 85.9%.	(Guo et al. 2018)
2	Oxytetracycline; Tetracycline; Sulfadiazine; Sulfamethazine	Photo-Fenton: UV/ pre-magnetized Fe0 /H2O2 process (UV/pre-Fe0 /H2O2)	Reaction time= 60 min. Static and uniform 200 mT magnetic field (MF) for 2 min was provided.	HPLC, TOC	With Fe0 /H2O2 and pre-Fe0 /H2O2 less than 10 and 20 % removal occurred of SMT while under UV-irradiation 68.6% SMT removal achieved in 60 min. also efficient removal of TC, OTC, and SD achieved.	(Pan et al., 2019)

Table 2: Effect of various biodegradation approaches on antibiotic at different experimental conditions

Order	Antibiotic	Biological strategy	Strain	Antibiotic Concentration	Temperature (°C) / pH	Duration (Time)	Removal efficie
1	Tetracycline	Bacterial remediation	*Klebsiella sp.*	61.27 mg/L	34.96°C/ 7.17	32 hours	Reduc TC wa 0.113
2	Enrofloxacin, Ceftiofur	Bacterial remediation	*Firmicutes, Proteobacteria, Actinobacteria, Bacteroidetes and Chloroflexi*	1 g/L	20°C	21 days	40-55, 100

3	Chlortetracycline, amoxicillin, sulfamethoxazole	Activated sludge	*Pseudmonas sp., Bacillus sp., Clostridium sp.*	2 mg/L	30°C/ 7.0	8 days	69.4, 8 89.6%
4	Ciprofloxacin	Bacterial remediation	*Microbial consortium (Achromobacter, Bacillus, Lactococcus, Ochrobactrum, and Enterococcus)*	5 mg/L	30°C	14 days	63
5	Streptomycin	Fungal remediation	*Ceriporia lacerata and Trametes versicolor*	100-400 ppm	25±2 °C	--	At 10 88.8% by Tr versic 79.1% Cerip lacera ppm.
6	Oxacillin, cloxacillin, and dicloxacillin	Fungal remediation	*Leptosphaerulina* sp.	16000 μg L^{-1}, 17500 μg L^{-1}, 19000 μg L^{-1}	25°C/ 5.6	15 days	100, 100, 81
7	Ciprofloxacin, ofloxacin	Fungal remediation	*T. harzanium and T. asperellum,*	400 mg/L	--	13 days	81 (C harza 40 (O strain
8	Azithromycin, erythromycin	Algal remediation	*C. pyrenoidosa*	385 ng/L, 661 ng/L	--	144 hours	48-10
9	Spiramycin, amoxicillin	Algal remediation	*Microcystis aeruginosa*	50 ng/L – 1 mg/L	--	7 days	12.5– 30.5–
10	Ciprofloxacin, Tetracycline,	Phytoremediation	*Chrysopogon zizanioides* (Vetiver grass)	10 mg/L	25 ± 3°C	30 days	>90
11	Ciprofloxacin	Phytoremediation	*Chrysopogon zizanioides* (Vetiver grass)	1 mg/L, 10 mg/L	25 ± 3°C	30 days	>80
12	Ciprofloxacin	Biochar based adsorption	Peels of pomelo (grapefruit)	10, 50 mg/L	20°C/ 3	--	Sorpti capaci 3.31

Graphical abstract



Highlights

- Degradation of recalcitrant antibiotics is an indispensable task
- Various physico-chemical and biological degradation processes are demonstrated
- AOPs: Fenton/ Photo-Fenton, Electrochemical, Photo catalysis are mostly utilized
- Role of microbial strain, bioreactor, biocatalysts, MFCs, and biochar is discussed
- Combined/ hybrid strategies are considered more efficient for effluent treatment